



Application of Lithium Organoborate with Salicylic Ligand to Lithium Battery Electrolyte

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The thermal characteristics of lithium bis[salicylato(2-)]borate (LBSB) and its novel derivatives synthesized by us, such as lithium bis[3-methylsalicylato(2-)]borate (3-MLBSB), lithium bis[3,5-dichlorosalicylato(2-)]borate (DCLBSB), and lithium bis[3,5,6-trichlorosalicylato(2-)]borate (TCLBSB) were examined by thermogravimetric analysis (TG). The thermal decomposition in air begins at 260, 290, 310, and 320°C for TCLBSB, LBSB, DCLBSB, and 3-MLBSB, respectively. The thermal stabilities of 3-MLBSB and DCLBSB are nearly equal to those of LiN(CF₃SO₂)₂ and LiN(C₂F₅SO₂)₂. The order of the stability toward oxidation of these organoborates is TCLBSB ≈ DCLBSB > LBSB > 3-MLBSB, which differs from the thermal stability. Ionic dissociation properties of LBSB and its derivatives were examined by conductivity measurements in ethylene carbonate-1,2-dimethoxyethane (EC-DME) equimolar binary dilute solutions. The conductivities of the 0.1 mol dm⁻³ DCLBSB and TCLBSB electrolytes become higher than those in the LBSB and 3-MLBSB electrolytes. It means that DCLBSB and TCLBSB have high dissociating abilities in EC-DME mixture. The 0.5 mol dm⁻³ LBSB/EC-DME equimolar binary solution exhibits the highest lithium electrode cycling efficiency of more than 85% in the higher range of cycle numbers. This is a good electrolyte for rechargeable batteries.

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Many workers have attempted to develop improved electrolytes for lithium batteries with high energy density and good rechargeability as power sources for many portable instruments and electric vehicles in order to protect the natural environment and resources. Recently, Barthel *et al.*¹⁻⁵ reported a new class of electrochemically and thermally stable lithium salts with a chelate-type anion of boron, such as lithium bis[1,2-benzenediolato(2-)-O,O']borate (LBBB), lithium bis[2,3-naphthalenediolato(2-)-O,O']borate (LBNB), lithium bis[2,2'-biphenyldiolato(2-)-O,O']borate (LBBPB), and lithium bis[salicylato(2-)]borate (LBSB) as shown in Fig. 1. In previous papers,⁶⁻⁸ we reported the thermal characteristics of LBBB, LBNB, and LBBPB and the electrolytic behavior in propylene carbonate (PC)-1,2-dimethoxyethane (DME), ethylene carbonate (EC)-DME, PC-tetrahydrofuran (THF), and EC-THF binary solvent electrolytes containing LBBB, LBNB, and LBBPB.

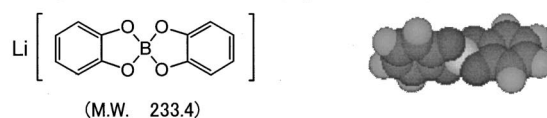
In this paper, the thermal characteristics of LBSB and its novel derivatives synthesized by us, such as lithium bis[3-methylsalicylato(2-)]borate (3-MLBSB), lithium bis[3,5-dichlorosalicylato(2-)]borate (DCLBSB), and lithium bis[3,5,6-trichlorosalicylato(2-)]borate (TCLBSB) as shown in Fig. 2, were examined. In addition, the electrolytic behavior and charge-discharge characteristics of a lithium electrode in EC-DME binary solvent electrolytes containing LBSB, 3-MLBSB, DCLBSB, and TCLBSB were studied and compared with those in the LiPF₆ electrolyte, which is a common salt for lithium batteries.

Experimental

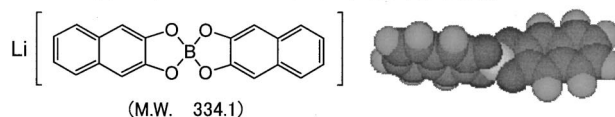
Materials.—The LBSB raw salt was easily obtained using lithium carbonate as the starting material instead of lithium hydroxide monohydrate of Barthel *et al.*⁴ The lithium salt was purified several times by recrystallization from acetonitrile and dried *in vacuo* at 120°C for 24 h. The yield was about 40%. The compounds of 3-MLBSB, DCLBSB, and TCLBSB were directly obtained from the reaction of Li₂CO₃, H₃BO₃, and three groups of salicylic acids, such as 3-methylsalicylic, 3,5-dichlorosalicylic, and 3,5,6-trichlorosalicylic acids. The 3-MLBSB and TCLBSB raw salts were recrystallized four times from acetonitrile. The DCLBSB raw salt was recrystallized four times from THF. The yields of these lithium

salts were about 30%. The purity and structure were confirmed by elemental and inductively coupled plasma analyses for B and Li, and C-H correlation spectroscopy (COSY), and ¹¹B nuclear magnetic resonance (NMR). The C-H COSY and ¹¹B NMR spectra of 3-MLBSB, for example, are shown in Fig. 3 and 4, respectively.

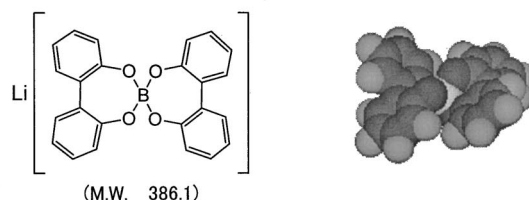
Lithium bis[1,2-benzenediolato(2-)-O,O']borate (LBBB)



Lithium bis[2,3-naphthalenediolato(2-)-O,O']borate (LBNB)



Lithium bis[2,2'-biphenyldiolato(2-)-O,O']borate (LBBPB)



Lithium bis[salicylato(2-)]borate (LBSB)

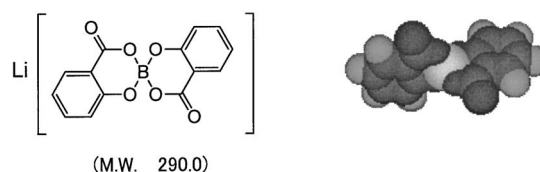
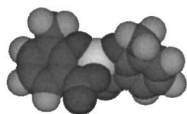
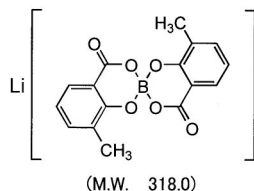


Figure 1. Structures of LBBB, LBNB, LBBPB, and LBSB.

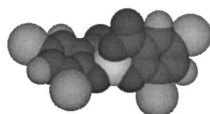
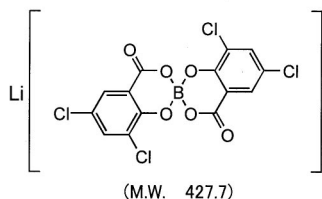
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Lithium bis[3-methylsalicylato(2-)]borate (3-MLBSB)



Lithium bis[3, 5-dichlorosalicylato(2-)]borate (DCLBSB)



Lithium bis[3,5,6-trichlorosalicylato(2-)]borate (TCLBSB)

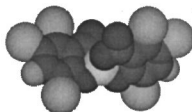
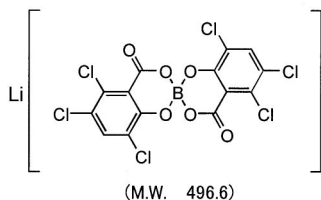
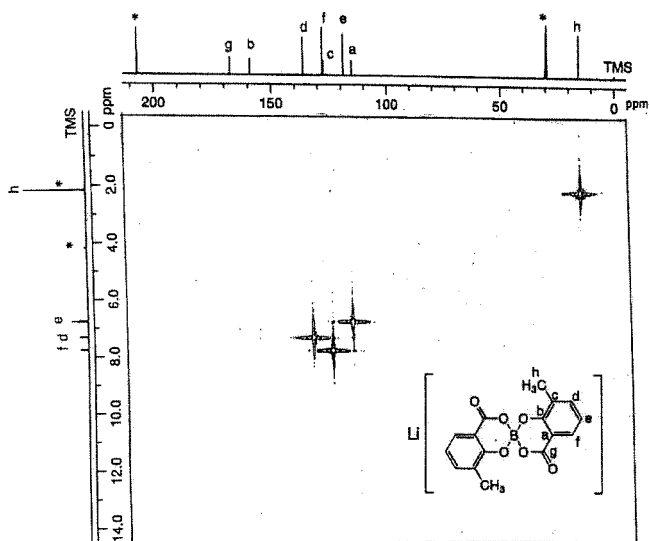
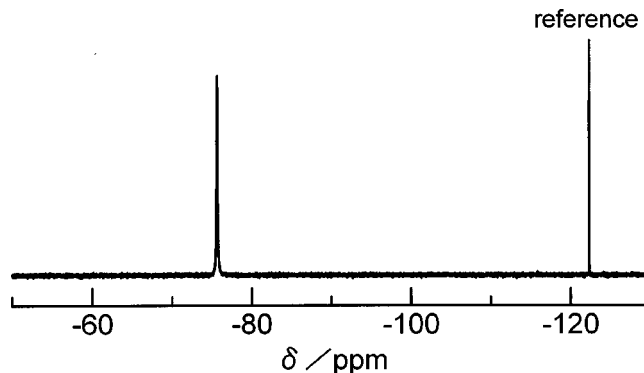


Figure 2. Structures of LBSB derivatives.

Figure 3. C-H COSY NMR spectra of 3-MLBSB in $(\text{CD}_3)_2\text{CO}$ with 1% tetramethyl silane. (*) Peak due to $(\text{CD}_3)_2\text{CO}$.

Commercial reagent grade DME was purified by refluxing overnight with LiAlH_4 , followed by fractional distillation.⁹ The solvent was dehydrated using purified molecular sieves (4 Å). EC was used as received (battery grade). Commercial reagent grade of LiPF_6 was dried *in vacuo* for 24 h at 70°C.

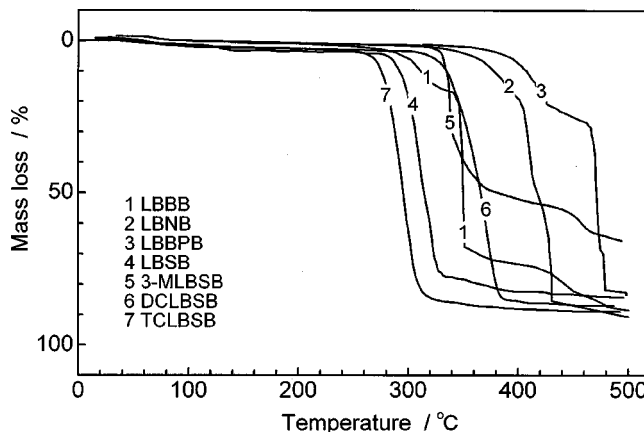
Apparatus and measurements.—Thermogravimetric (TG) analyses of the lithium organoborates were carried out with equipment from Seiko Instruments [TG/differential thermal analysis (DTA) 220] using a sample mass of about 10 mg. The NMR spectra were measured with JEOL JNM-LA500. The conductivities of the electrolytes were measured using a conductometer (Toa Electronics,

Figure 4. ^{11}B NMR spectra of 3-MLBSB in $(\text{CD}_3)_2\text{CO}$. External reference: $0.1 \text{ mol dm}^{-3} \text{ NaBH}_4$ in D_2O .

model CM-60S) with a cell (model CGT-511B). The decomposition voltages (i-E curves) of the electrolytes using a three-electrode system (Pt plate working, Li plate counter, and Li plate reference electrodes) were measured at a scan rate of 10 mV s^{-1} with a potentiostat (Hokuto Denko, model HA-501) equipped with a potential sweep unit (model HB-104). The Pt electrode was polished with aluminum oxide (particle size, *ca.* 0.05 μm). The lithium electrode cycling efficiency (charge-discharge coulombic efficiency of lithium electrode) was estimated using a six-channel charge/discharge unit (Hokuto Denko, model HJ-101SM6) by a galvanostatic plating/stripping method which was reported by Koch and Brummer.¹⁰ Preparation of the electrolyte solutions and the cell assembly were carried out in an argon-filled dry box. The apparatus and techniques for measuring the viscosity in electrolyte solutions were similar to those previously used.¹¹

Results and Discussion

Thermal stability.—Figure 5 shows the TG curves of the lithium organoborates in air. In LBSB and its derivatives, 3-MLBSB exhibits a higher thermal stability. TCLBSB (trichloro compound) with heavier molecule shows the lowest thermal stability. This means that the trichloro compound such as TCLBSB in spite of having a heavier molecule, decomposes easily on heating in air at lower temperature. The decomposition started to occur at 260, 290, 310, and 320°C for TCLBSB, LBSB, DCLBSB, and 3-MLBSB, respectively. Accordingly, it was found that the thermal stability depends on the inductive effects of the substituent groups to the chelate-type anion with boron. In addition, the lower thermal stability of TCLBSB compared to DCLBSB seems to be dependent on the structural steric

Figure 5. TG curves of lithium organoborates at a heating rate of 1°C min^{-1} in air. α -Alumina was used as a reference.

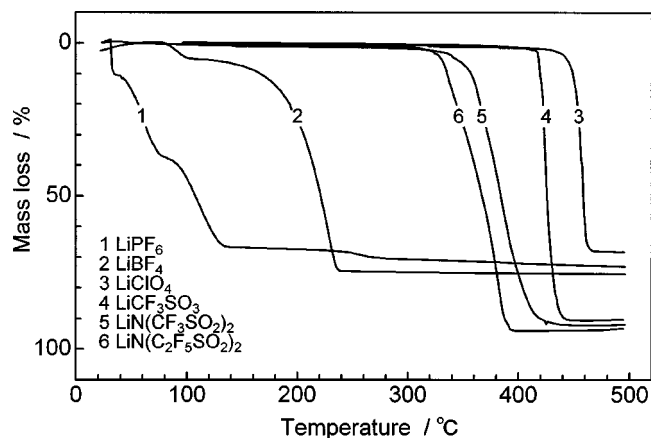


Figure 6. TG curves of lithium salts at a heating rate of 1°C min^{-1} in air. α -Alumina was used as a reference.

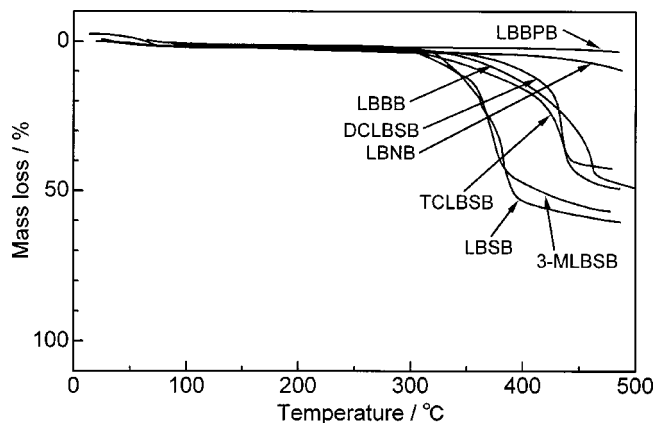


Figure 7. TG curves of lithium organoborates at a heating rate of 1°C min^{-1} in N_2 gas. α -Alumina was used as a reference.

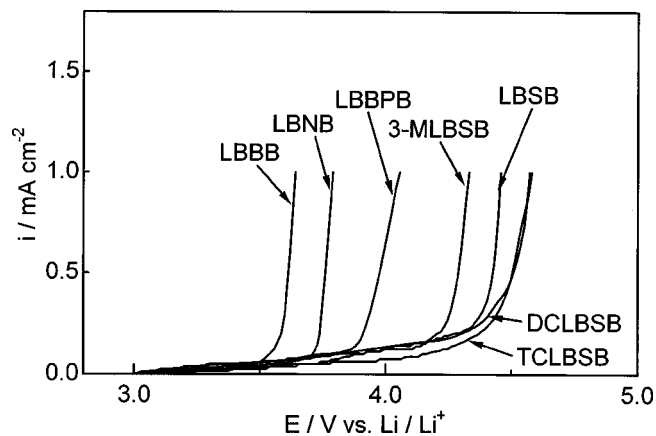


Figure 8. i-E curves in PC solutions containing 0.1 mol dm^{-3} electrolytes using a Pt working electrode at a scan rate of 10 mV s^{-1} at 25°C .

instability. Furthermore, the thermal stability of 3-MLBSB and DCLBSB is nearly equal to that of the fluorinated lithium salts in Fig. 6, such as $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ and $\text{LiN}(\text{C}_2\text{F}_5\text{SO}_2)_2$, which are considered possible lithium battery electrolytes. Figure 7 shows the TG curves of the lithium organoborates in N_2 gas. The thermal stabilities of LBBPB and LBNB are very high. Especially, LBBPB scarcely decomposes after 8 h. The high thermal stabilities of LBBPB and LBNB are dependent on the higher conjugate energies of naphtha-

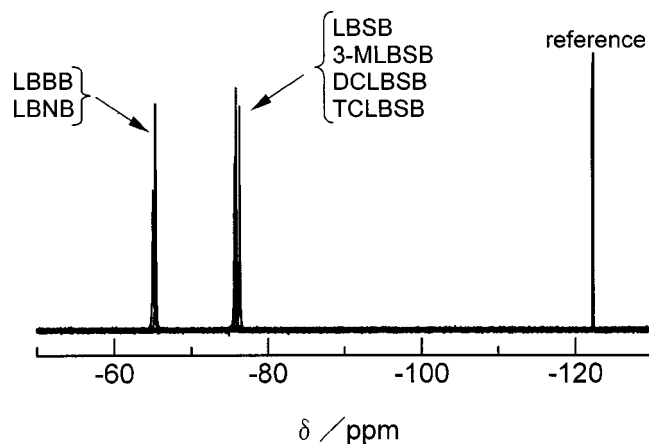


Figure 9. ^{11}B NMR spectra of lithium organoborates in $(\text{CD}_3)_2\text{CO}$. External reference: $0.1 \text{ mol dm}^{-3} \text{ NaBH}_4$ in D_2O .

lene and biphenyl. The stability of the TCLBSB becomes much higher than that of LBSB and 3-MLBSB in N_2 gas. In addition, 3-MLBSB has lower stability in N_2 gas compared with that in air. This means that the substitution of methyl group is effective in depressing the oxidative decomposition in air.

Decomposition voltage (i-E curve).—The i-E curves of the PC solution containing 0.1 mol dm^{-3} lithium organoborates are shown in Fig. 8. The order of the oxidation stability in these organoborates is $\text{TCLBSB} \approx \text{DCLBSB} > \text{LBSB} > \text{3-MLBSB} > \text{LBBPB} > \text{LBNB} > \text{LBBB}$, which is different from that of the

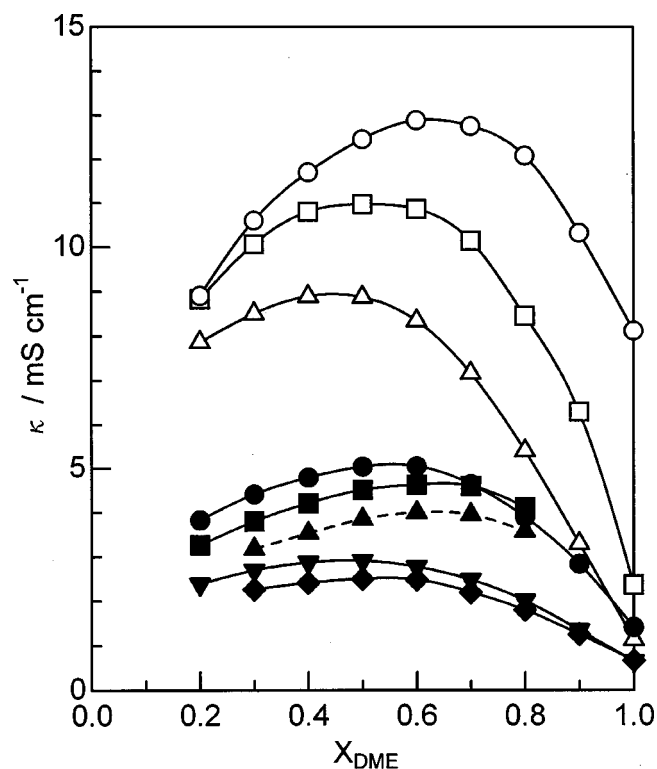


Figure 10. Variation in specific conductivities (κ) of EC-DME solutions containing 0.5 mol dm^{-3} electrolytes at 25°C . X_{DME} denotes the mole fraction of DME. Concentration of DCLBSB: 0.3 mol dm^{-3} . ($\square$) LiClO_4 , ($\circ$) LiPF_6 , ($\triangle$) LiBF_4 , ($\bullet$) LBBB, ($\blacksquare$) LBNB, ($\blacktriangledown$) LBSB, ($\blacklozenge$) 3-MLBSB, ($\blacktriangle$) DCLBSB.

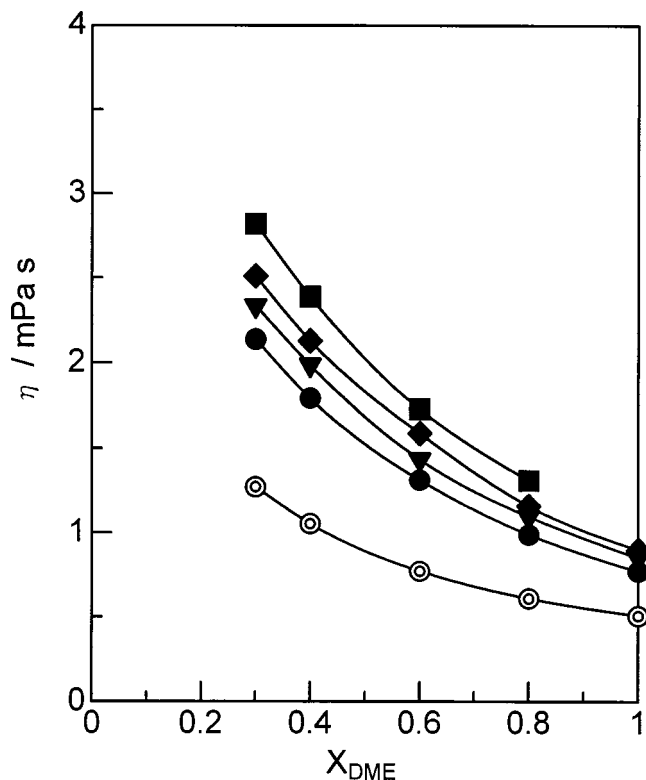


Figure 11. Variation in viscosities (η) of EC-DME solutions containing 0.5 mol dm⁻³ electrolytes at 25°C. X_{DME} denotes the mole fraction of DME. (●) LBBB, (■) LBNB, (▼) LBSB, (◆) 3-MLBSB, (○) solvent.

thermal stability in Fig. 5. However, LBBB shows not only the lowest thermal stability, but also lowest electrochemical stability. On the other hand, a high oxidation stability of TCLBSB and DCLBSB is dependent on the high charge delocalization around the boron in the chelate-type anion due to the electron withdrawing by the chlorine atoms. Figure 9 shows the ¹¹B NMR spectra of the lithium organoborates except for LBBPB. It is difficult for LBBPB to dissolve in (CD₃)₂CO. The differences of the chemical shifts (δ) of LBBB or LBNB, and of LBSB or its derivatives are within 0.5 and 0.7 ppm, respectively. Accordingly, it is impossible from the chemical shifts of ¹¹B NMR spectra to explain in detail the oxidation stability for lithium organoborates in Fig. 8. There is no proportionality between the ¹¹B NMR chemical shift and the oxidation potential of these organoborates. It means that there is a clear delineation between the chelates containing only phenolic oxygens (easily oxidized) and those where a carboxylic group is involved (oxidized beyond 4 V). However, it is found that both spectra of LBBB and LBNB are shifted downfield to a (low electron density) of more than 10 ppm compared with those of LBSB and its derivatives. Therefore, it is difficult from ¹¹B NMR spectra to explain higher oxidation stabilities of LBSB and its derivatives than those of LBBB and LBNB. It seems that the oxidative decomposition mechanisms for LBBB and LBNB are different from those for LBSB and its derivatives.

Conductivity.—The variation in specific conductivities (κ) in EC-DME solutions containing 0.5 mol dm⁻³ electrolytes at 25°C is shown in Fig. 10. It is difficult to dissolve 0.5 mol dm⁻³ DCLBSB in an EC-DME mixture. As can be seen from Fig. 11, the conductivities in the lithium organoborate solutions except for LBNB vary inversely with the viscosities of the solutions. The ionic dissociation of LBNB with a higher molecular weight than that of LBSB and 3-MLBSB becomes greater in the EC-DME mixture. The conducting behavior of LBNB is different from that of the other lithium

Table I. Specific conductivities in EC-DME equimolar binary solutions containing 0.1, 0.3, and 0.5 mol dm⁻³ LBSB and its derivatives at 25°C.

mol dm ⁻³	Conductivity (mS cm ⁻¹)		
	0.1	0.3	0.5
LBSB	1.77	2.82	3.04
3-MLBSB	1.53	2.43	2.50
DCLBSB	2.31	3.85	Partially soluble
TCLBSB	2.33	Partially soluble	Partially soluble

organoborates, similarly in PC and EC-based THF and DME binary solvent mixtures as previously reported.⁸ Ionic dissociation properties of LBSB and its derivatives were examined by conductivity measurements in dilute solutions. The specific conductivities for different concentrations of these lithium organoborates in EC-DME equimolar binary solutions are shown in Table I. The conductivities of the 0.1 mol dm⁻³ DCLBSB and TCLBSB electrolytes, which have higher molecular weights and larger anions, become higher than those in the LBSB and 3-MLBSB electrolytes. This means that DCLBSB and TCLBSB are apparently highly dissociating in EC-DME mixture. In addition, the higher dissociating abilities are influenced by the high charge delocalization of the 3,5-dichlorosalicylate and 3,5,6-trichlorosalicylate ions.

Lithium electrode cycling efficiency.—Figure 12 shows the lithium electrode cycling efficiencies (charge/discharge coulombic efficiency of the lithium electrode). The 0.5 mol dm⁻³ LBSB electrolyte exhibits the highest efficiency, viz., more than 85% in a higher range of cycle numbers. In addition, the 3-MLBSB and DCLBSB electrolytes show a moderate efficiency for a higher range of cycle numbers, whereas the LiPF₆ electrolyte shows an efficiency decreasing with cycle number. By comparing the cycling efficiencies in Fig. 12 and the conductivities in Fig. 10, it is found that the efficiency is not dependent on the conductivity of the electrolytes. This means that the efficiency depends on the morphology, chemical composition and thickness of the passivating films formed on the electrode due to the reduction of reactive impurities and electrolytes.

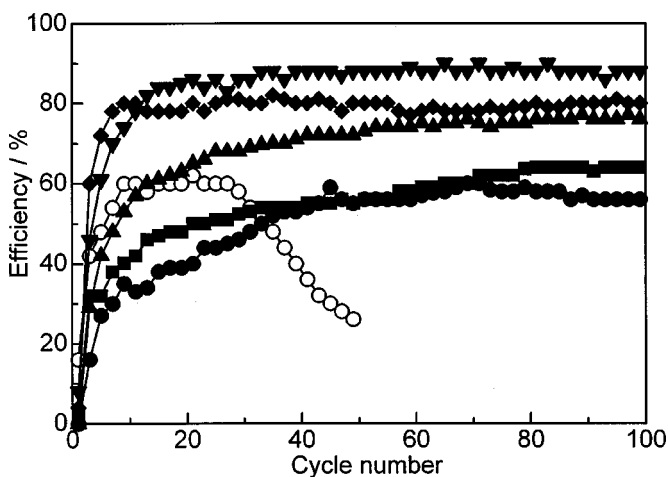


Figure 12. Lithium electrode cycling efficiencies in EC-DME equimolar binary solutions containing 0.5 mol dm⁻³ LiPF₆ and lithium organoborates at 25°C. Concentration of DCLBSB: 0.3 mol dm⁻³. $i_p = i_s = 0.1$ mA cm⁻². $Q_p = 0.01$ C cm⁻². (○) LiPF₆, (●) LBBB, (■) LBNB, (▼) LBSB, (◆) 3-MLBSB, (▲) DCLBSB.

Conclusion

The 0.5 mol dm⁻³ LBSB/EC-DME equimolar binary solution showing the highest cycling efficiency of more than 85% at a higher range of cycle number is a good electrolyte for rechargeable batteries. In addition, the 3-MLBSB and DCLBSB solutions show moderate cycling efficiencies.

Acknowledgments

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